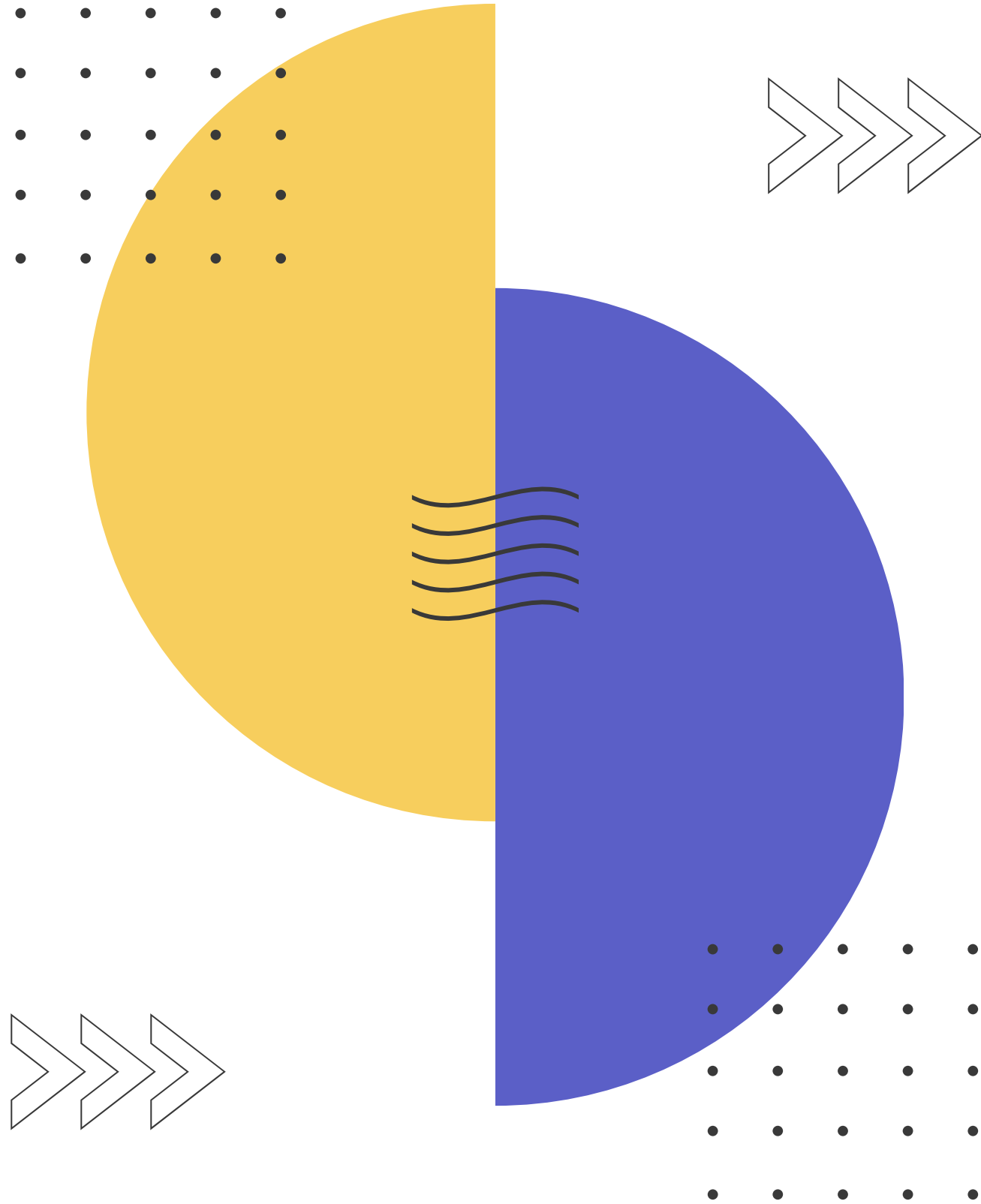
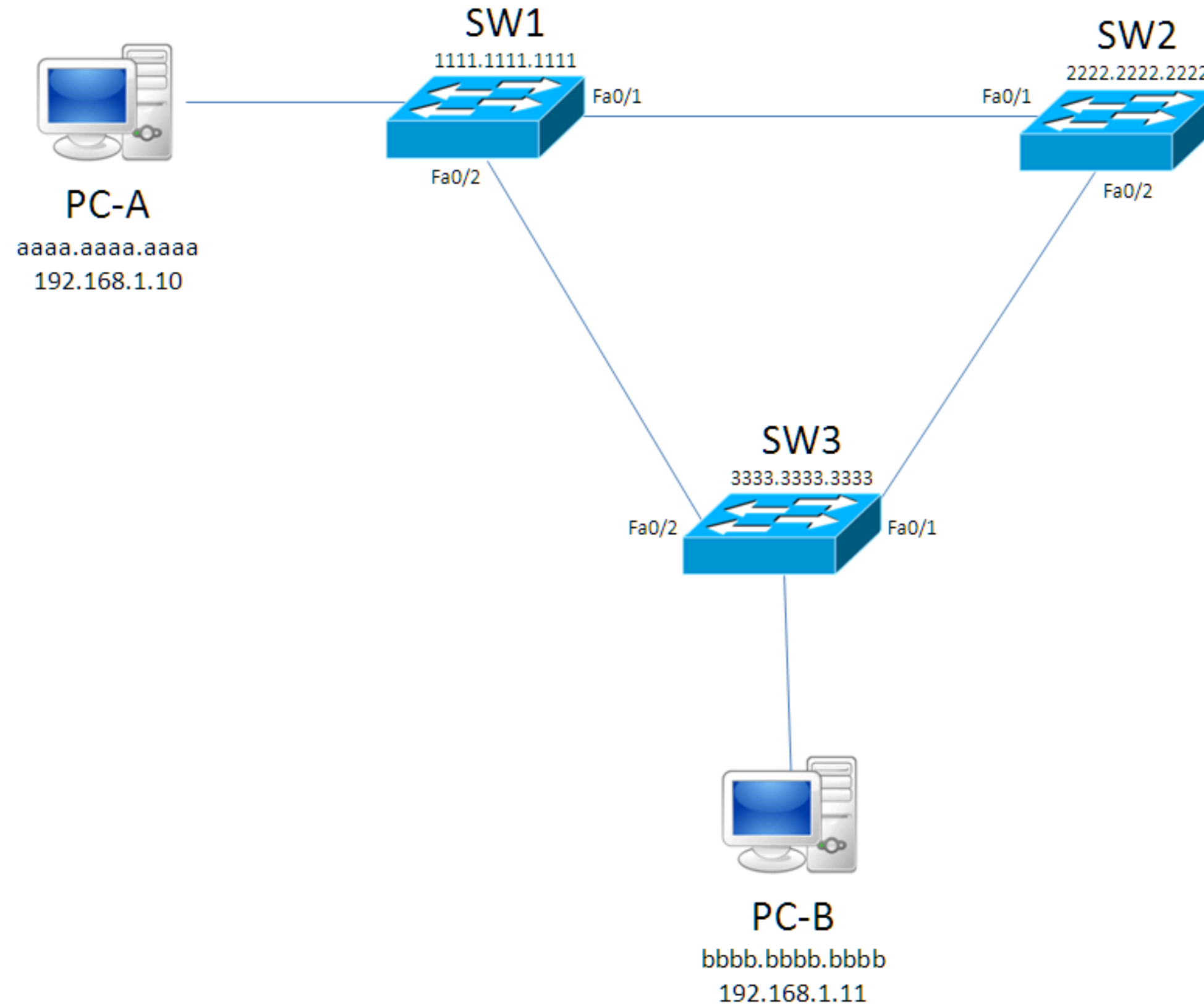




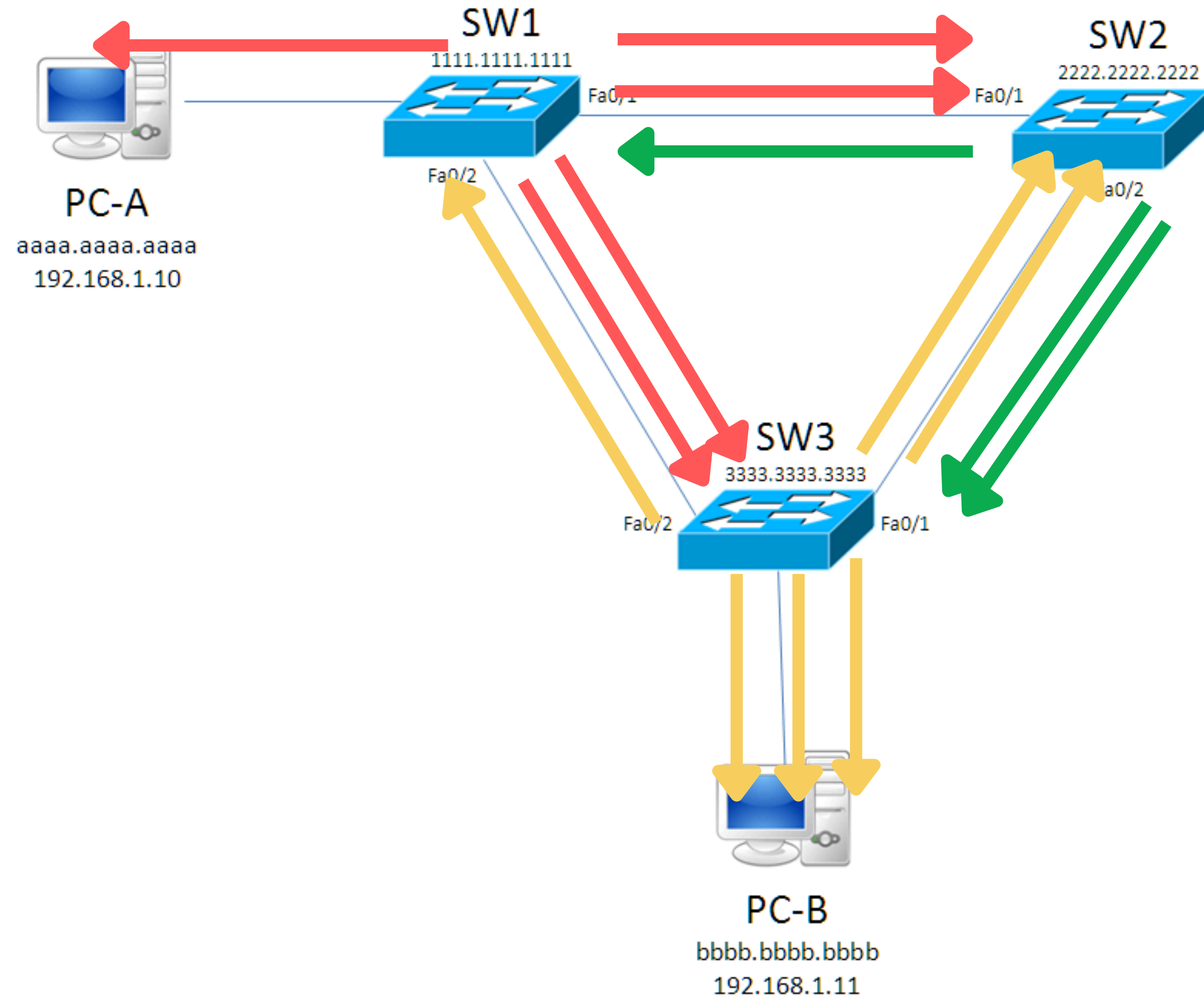
IT INFRASTRUCTURE AND COMPUTER NETWORKS

Week 5

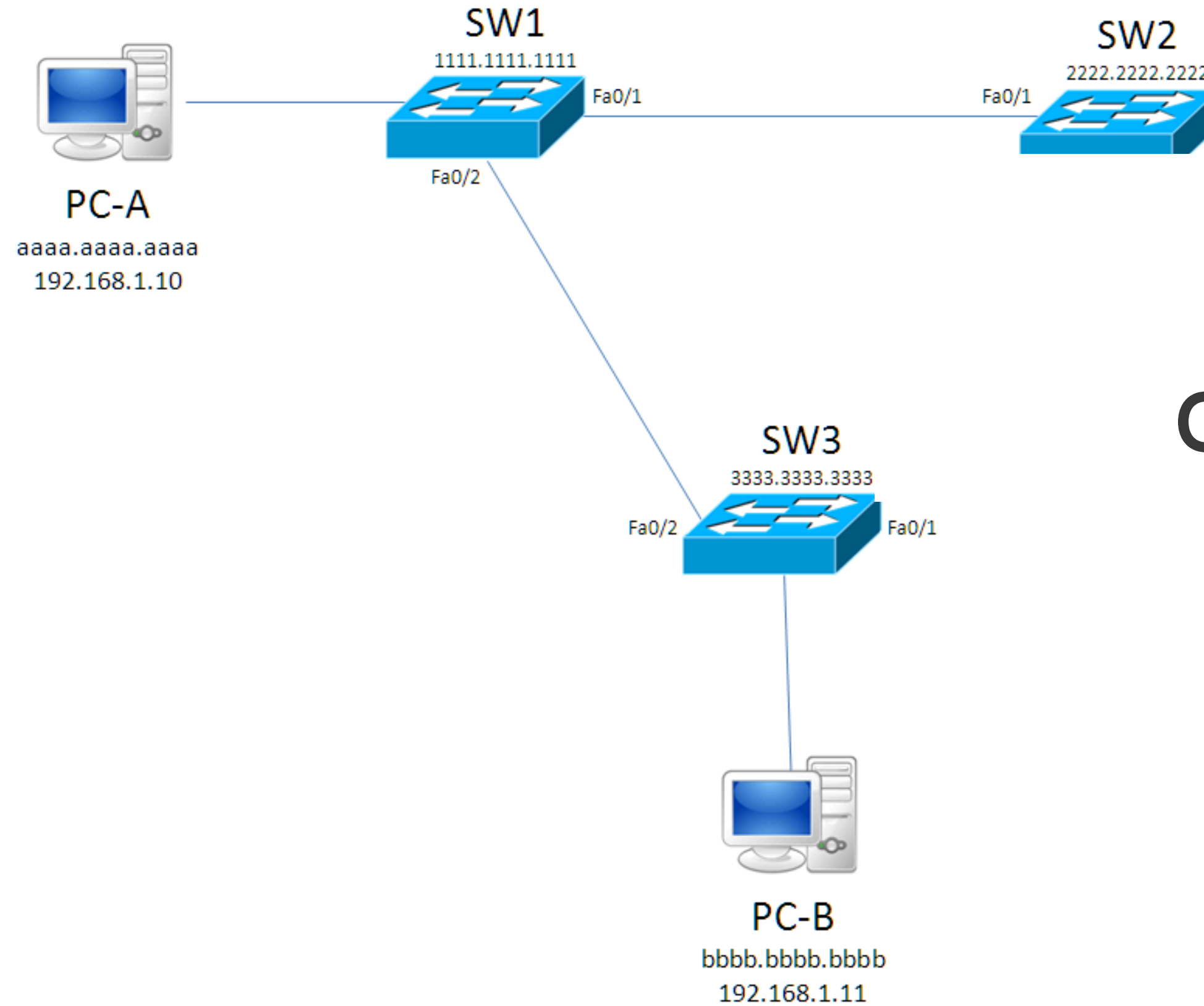
WHAT IS THE PROBLEM WITH THIS TOPOLOGY?



LOOPS!

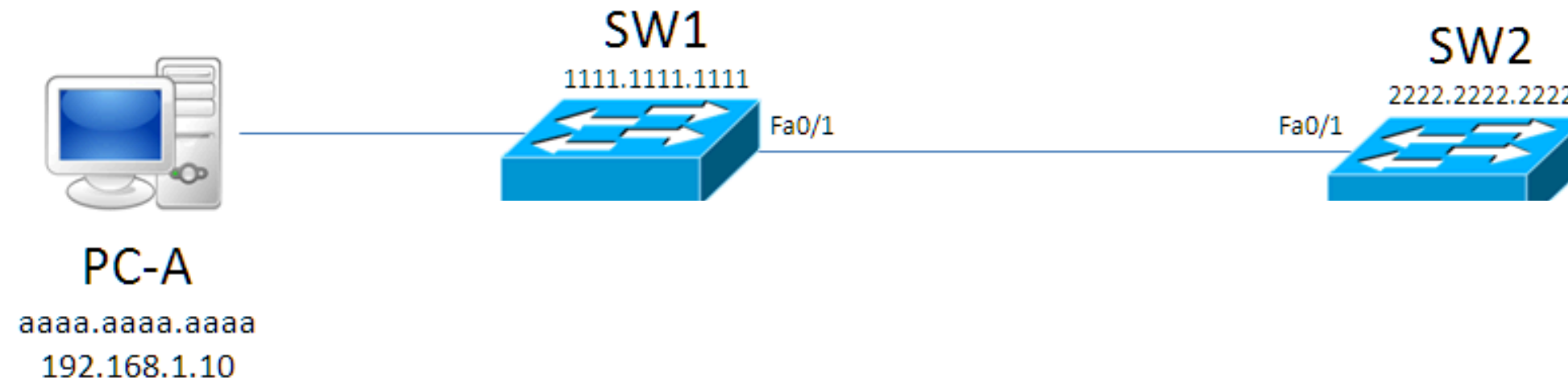


HOW TO SOLVE IT?



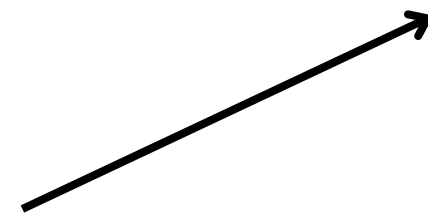
**REMOVE
CONNECTION?**

HOW TO SOLVE IT?

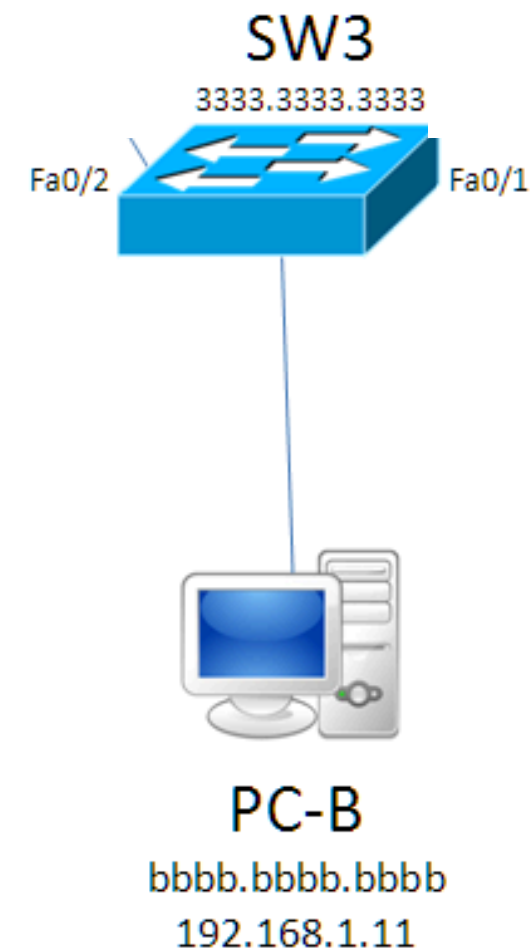


BAD IDEA

if this connection is lost
too, SW3 will disconnect
from network



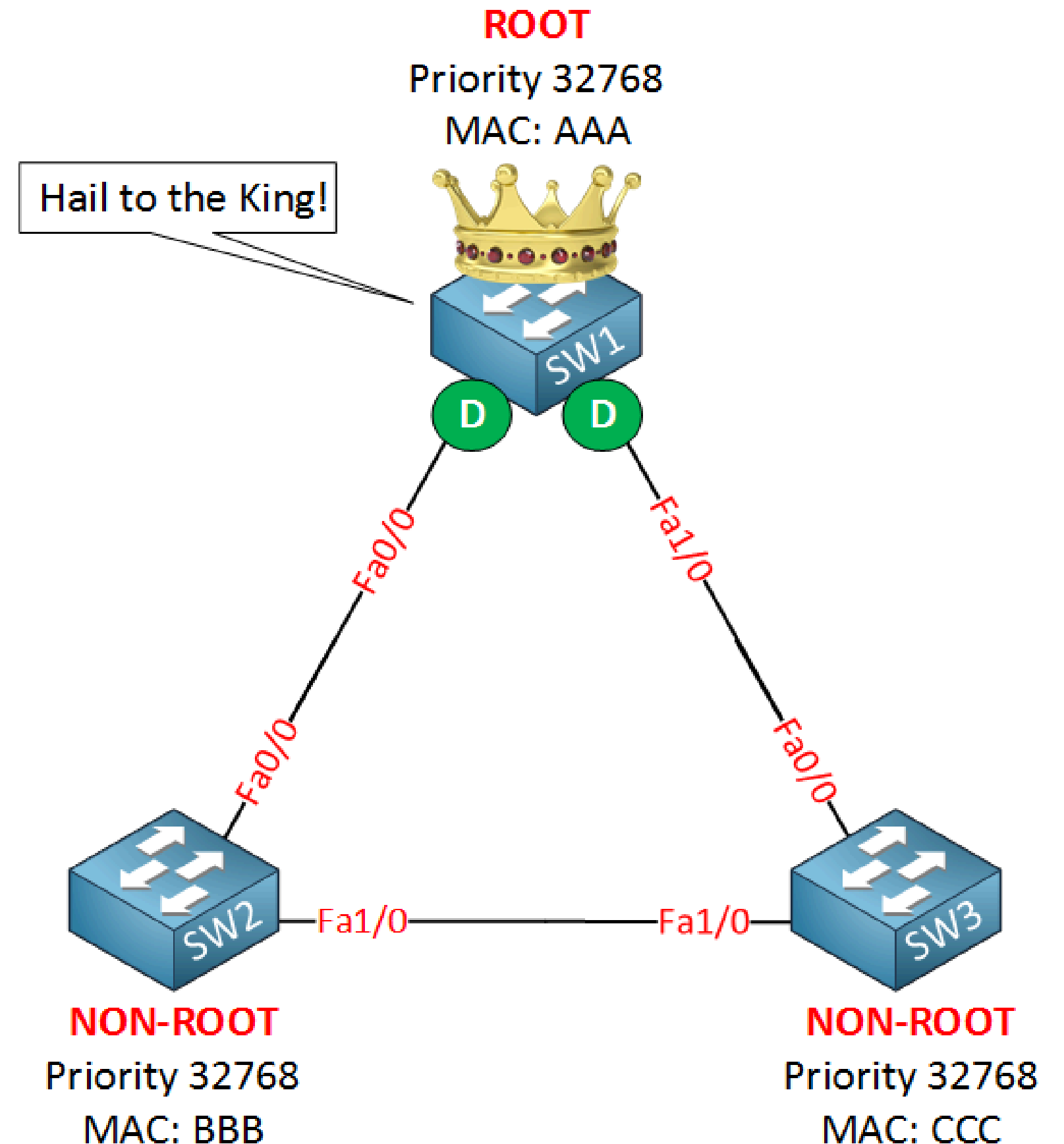
**REMOVE
CONNECTION?**



STP

Spanning Tree Protocol

- Prevents loops in Layer 2 networks
 - Creates a loop-free logical topology
 - Blocks redundant paths
-
- One switch (with lowest Bridge ID) becomes Root Bridge
 - All path calculations are based on the Root Bridge

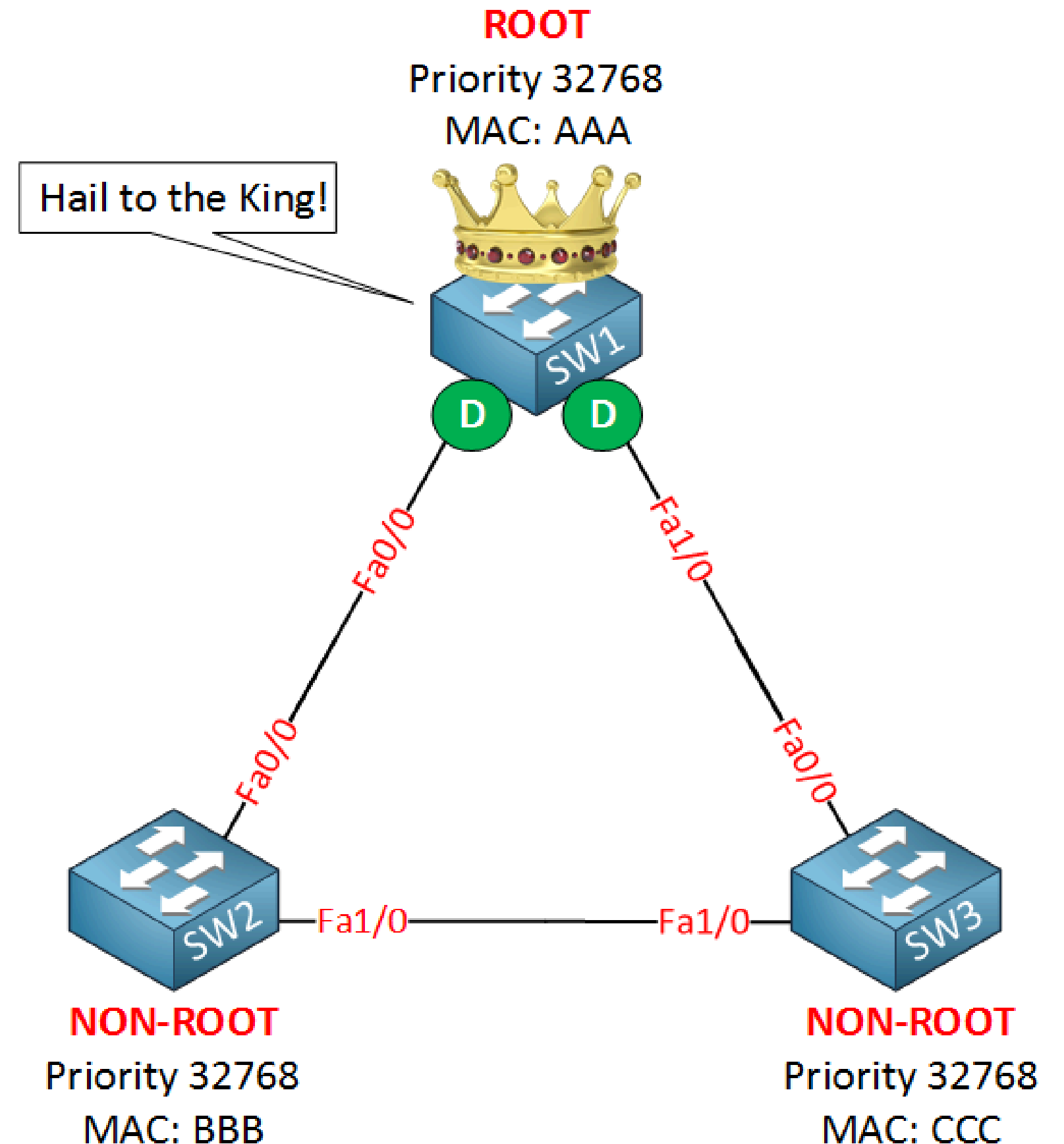


PORT ROLES IN STP

- Root Port (RP) – best path to root
- Designated Port (DP) – forwards traffic
- Blocked Port – prevents loops

PORT STATES

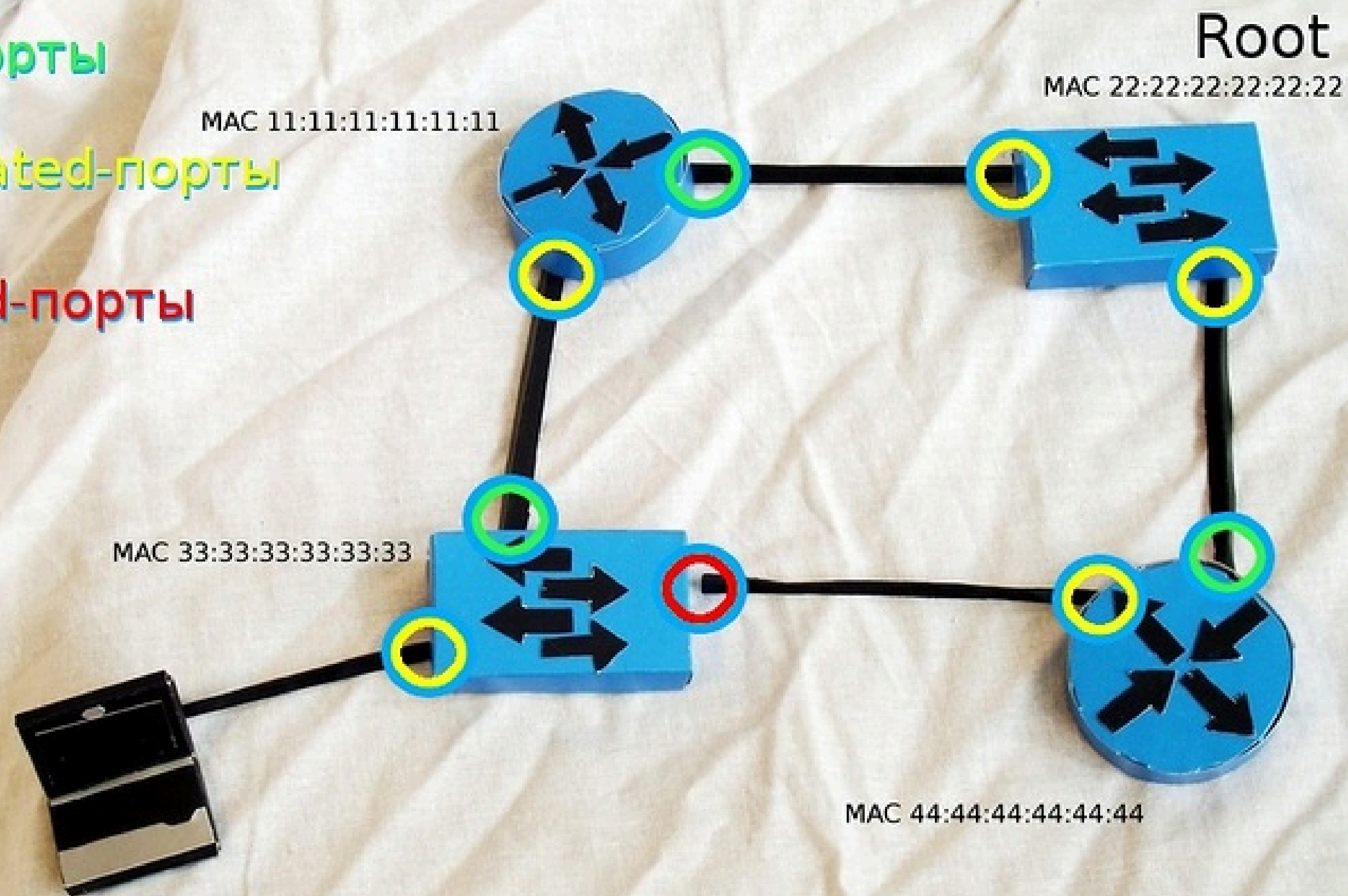
- Blocking
- Listening
- Learning
- Forwarding
- Disabled



○ Root-порты

○ Designated-порты

○ Blocked-порты



TYPES OF STP

- RSTP — Rapid STP — Fast
- MSTP — Multiple STP — For big networks
- PVST (+RPVST) — Per-Vlan — Proprietary Cisco protocol

CONFIGURATION

097, y: 77


2960-24TT
Switch0

 Switch0

IOS Command Line Interface

Compiled Wed 26-Jun-13 02:49 by mnguyen

Press RETURN to get started!

Switch>en

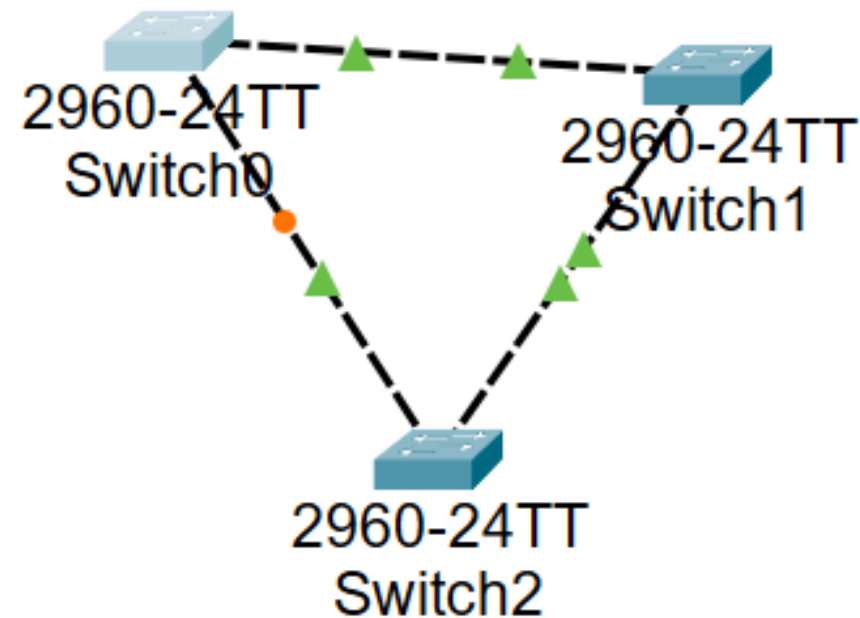
Switch#show spa

Switch#show spanning-tree

No spanning tree instance exists.

Switch#|

CONFIGURATION



```
Switch#show spanning-tree
```

```
VLAN0001
```

```
Spanning tree enabled protocol ieee
```

```
Root ID      Priority      32769
              Address      0002.17E2.5E37
              Cost        19
              Port        1(FastEthernet0/1)
              Hello Time  2 sec    Max Age 20 sec    Forward Delay 15 sec
```

```
Bridge ID    Priority      32769  (priority 32768 sys-id-ext 1)
              Address      0090.2142.B6C7
              Hello Time  2 sec    Max Age 20 sec    Forward Delay 15 sec
              Aging Time  20
```

Interface	Role	Sts	Cost	Prio.Nbr	Type

Fa0/1	Root	FWD	19	128.1	P2p
Fa0/2	Altn	BLK	19	128.2	P2p

CONFIGURATION

```
Switch(config)#spanning-tree mode rapid-pvst
```

```
Switch(config)#end
```

```
Switch#
```

```
%SYS-5-CONFIG_I: Configured from console by console
```

```
Switch#show spanning-tree
```

```
VLAN0001
```

```
Spanning tree enabled protocol rstp
```

```
Root ID    Priority    32769
           Address    0002.17E2.5E37
           Cost      19
           Port      1(FastEthernet0/1)
           Hello Time 2 sec  Max Age 20 sec  Forward Delay 15 sec
```

```
Bridge ID  Priority    32769  (priority 32768 sys-id-ext 1)
           Address    0090.2142.B6C7
           Hello Time 2 sec  Max Age 20 sec  Forward Delay 15 sec
           Aging Time 20
```

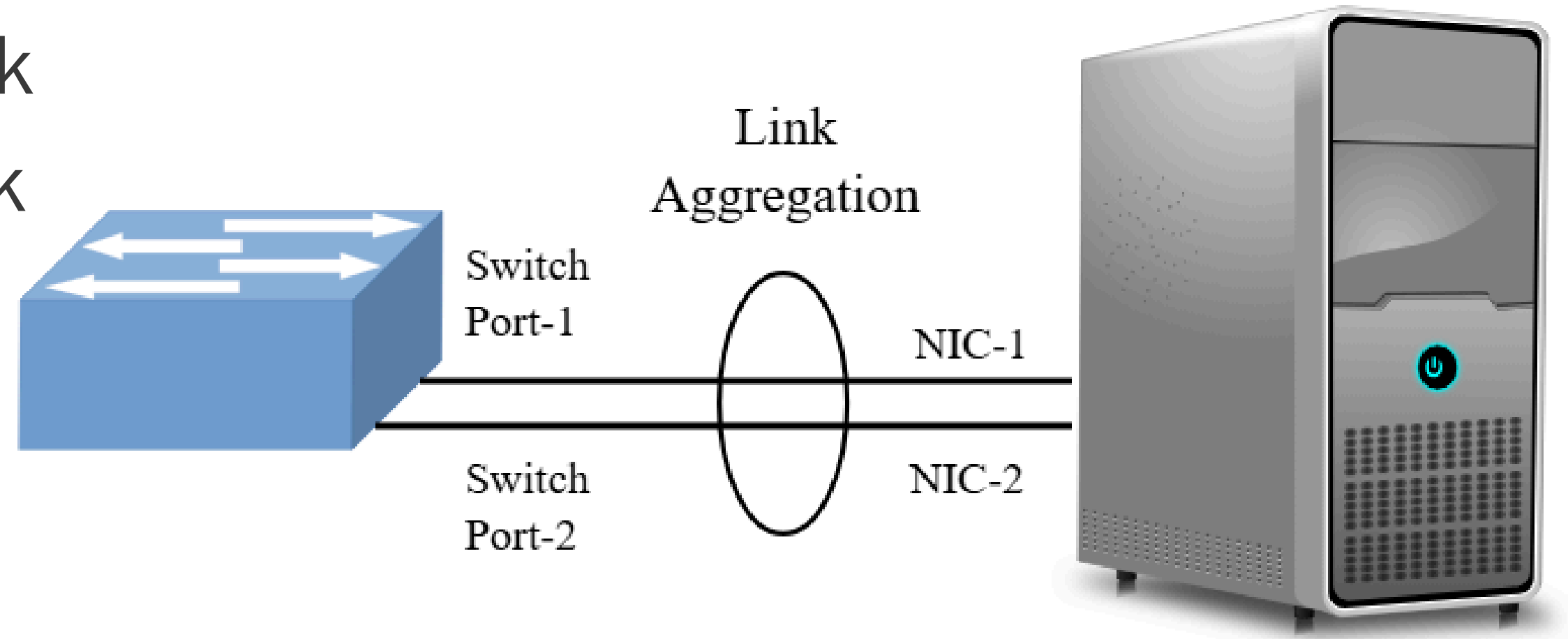
Interface	Role	Sts	Cost	Prio.Nbr	Type
Fa0/1	Root	LRN	19	128.1	P2p
Fa0/2	Altn	BLK	19	128.2	P2p

```
Switch#
```

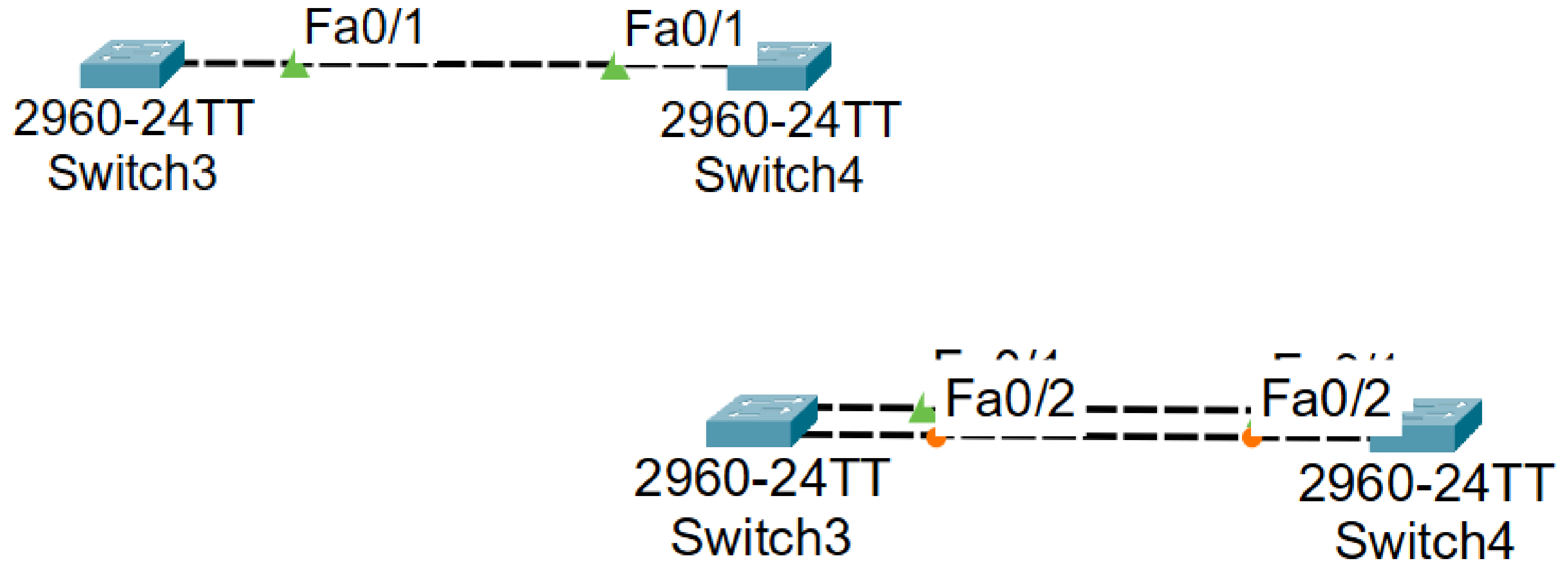
**IS IT THE ONLY WAY TO CREATE FAIL-SAFE
NETWORK?**

LINK AGGREGATION

- Combines multiple physical links
- Creates one logical link
- STP sees it as ONE link
- Increases bandwidth
- Provides redundancy



CONFIGURATION



CONFIGURATION

```
Switch(config)#interface range fa0/1-2
```

```
Switch(config-if-range)#channel-group 1 mode on
```

```
Switch(config-if-range)#
```

```
Creating a port-channel interface Port-channel 1
```

```
%LINK-5-CHANGED: Interface Port-channel1, changed state to up
```

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel1, changed  
state to up
```

```
Switch(config-if-range)#
```


CONFIGURATION

```
Switch#show etherchannel summary
```

```
Flags:  D - down          P - in port-channel
        I - stand-alone  s - suspended
        H - Hot-standby  (LACP only)
        R - Layer3       S - Layer2
        U - in use       f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port
```

```
Number of channel-groups in use: 1
```

```
Number of aggregators: 1
```

```
Group  Port-channel  Protocol  Ports
```

```
-----+-----+-----+-----
```

```
1      Po1 (SU)      -      Fa0/1 (P) Fa0/2 (P)
```

```
Switch#
```

CONFIGURATION

```
Switch(config)#interface port-channel 1
Switch(config-if)#sw
Switch(config-if)#switchport mode tr
Switch(config-if)#switchport mode trunk

Switch(config-if)#swi
Switch(config-if)#switchport tr
Switch(config-if)#switchport trunk allo
Switch(config-if)#switchport trunk allowed vlan 20,30
Switch(config-if)#
```

Any questions?

